Economic Analysis and Policy 68 (2020) 101–113

Contents lists available at ScienceDirect

Economic Analysis and Policy

journal homepage: www.elsevier.com/locate/eap

Policy debates and controversies

Wage and labor mobility between public, formal private and informal private sectors in a developing country

Mostafa E. Shahen [a,b], Koji Kotani [a,c,d,e,*], Makoto Kakinaka [e,f], Shunsuke Managi [d,g]

[a] School of Economics and Management, Kochi University of Technology, Japan
[b] Faculty of Commerce, Zagazig University, Egypt
[c] Research Institute for Future Design, Kochi University of Technology, Japan
[d] Urban Institute, Kyusyu University, Japan
[e] College of Business, Rikkyo University, Japan
[f] Graduate School for International Development and Cooperation, Hiroshima University, Japan
[g] Departments of Urban and Environmental Engineering, School of Engineering, Kyushu University, Japan

ARTICLE INFO

Article history:
Received 18 December 2019
Received in revised form 4 June 2020
Accepted 11 September 2020
Available online 15 September 2020

JEL classification:
J21
J23
J24
J31
O17

Keywords:
Wage gap
Public sector
Formal private sector
Informal private sector
Oaxaca–Blinder method
Difference-in-difference

ABSTRACT

Labor and wage in the public and private sectors are important factors in economies. In developing countries, the private sector is divided into formal and informal private sectors. Little research has addressed the changes in wage and labor among the public, formal private and informal private sectors over time within a single framework. We study the wage gap, labor mobility and the impact of changing employment sectors on wages by using the Oaxaca–Blinder decomposition and difference-in-difference (DID) methods with the Egyptian Labor Market Panel Survey data from 1998 to 2012. The decomposition shows that the wage gap between the public and formal (informal) private sectors has remained strong, with education, age and working experience being the driving forces. The DID method shows that the percentages and wage losses of movers from the formal private sector to the informal private sector are much higher and more significant than the percentages and wage losses of movers from the public sector to the informal private sector. In summary, Egyptian private sector employees face a high risk of unwillingly falling into and staying in the informal private sector, while the highly educated employees are only attracted to and stay long term in the public sector. These results demonstrate likely obstacles for further economic growth and the stability of the Egyptian economy and have implications for other developing and Arab countries with a sizeable public sector. The government may need to restructure wage systems, employment practices and cultures to strike a balance between the public and private sectors, providing people with incentive schemes and education to nurture (formalize) the formal (informal) private sector.

1. Introduction

The private sector is known to formally divide into the formal and informal ones, and it has been demonstrated that the formal private sector is a salient factor for the development of countries. In developed and OECD countries, the formal

* Corresponding author at: School of Economics and Management, Kochi University of Technology, Japan.
E-mail address: kojikotani757@gmail.com (K. Kotani).

https://doi.org/10.1016/j.eap.2020.09.006
0313-5926/

Nomenclature

DID	Difference-in-difference
ELMPS	Egyptian Labor Market Panel Survey
LE	Egyptian pound
OLS	Ordinary least squares

private sector is the main employer with 79% of working population on an average, while the rest is employed by the public sector (Hammouya, 1999). In other words, the percentage of employment in the informal private sector is negligible. In developing countries, the informal private sector employs more than 60% of working population, while the public and formal private sectors employ around 25% and 15%, respectively (Gërxhani, 2004; ILO News, 2018). Specifically, the formal private sector is usually anemic and faces many obstacles to grow, leading public and informal private sectors to remain the main employers in labor markets (Marinakis, 1994; ILO News, 2018). It is a big challenge for developing countries to have large informal, anemic formal private and medium public sectors in the labor markets for development as well as stability of the economy.

Governments in developing countries with an authoritarian regime, especially in Arab countries, use public sector employment as a bargaining chip to ensure the support and loyalty of people for decades (Assaad, 2014; Lindauer and Nunberg, 1996; Rama, 1997; Rodrik, 2000). However, governments cannot always sustain that level of public sector employment with an excess supply of labor within their countries. At the same time, the formal private sector, which is usually anemic and not well established, cannot absorb the excess supply in the labor market. This leads to the thriving of informal private sectors, which has negative consequences on economic and political stability, such as high tax evasion, low wages and insecure jobs (Meghir et al., 2015). To understand such a labor structure and its dynamics in developing countries, this study addresses the changes in wage gap and labor mobility among the public, formal and informal private sectors over time by taking the case of the Egyptian labor market.

Economists study the wage gap between the public and private sectors at the national and cross-national levels in developed countries. Papapetrou (2006) and San and Polat (2012) investigate the wage gap between public and private sectors, demonstrating that average wages in the private sector are lower than those in the public sector and the reasons why the public sector attracts employees in Greece and Turkey, respectively. Lucifora and Meurs (2006) analyze the pay gap between the private and public sectors in Italy, France and Great Britain. They find that low-skilled workers receive less pay in the private sectors than in the public sectors, while the opposite is true for high-skilled workers. In France and Italy, wages are not significantly different between the private and the public sectors. These studies show that the wage gap varies across countries and that the wage gap is one of the factors that attract individuals to one employment sector.

It is reported that the informal private sector thrives in most developing countries because of an inability of the public and formal private sectors to absorb the new entrants from the labor supply (Magnac, 1991; Maloney, 1999; Cunningham and Maloney, 2001; Maloney, 2004; Pratap and Quintin, 2006; Gunther and Launov, 2012). Although informal employment is considered to be a resort against unemployment in developing countries, it is associated with a pay penalty (Fields, 1975). Bargain and Kwenda (2014) investigate the wage gap between the formal and informal sectors in Brazil, Mexico and South Africa and find that the wages in the informal sectors are significantly lower than those in the formal sectors in these countries. Glinskaya and Lokshin (2007) examine the wage gap among the public, formal private and informal private sectors in India using cross-sectional data and point out that the public sector wage is the highest among the three sectors. These studies show that informal sector is accompanied by poor work conditions.

Some researchers study the labor market and employment in Egypt. Assaad (2014) examines the structure of the labor market and finds that informal employment increased over the period 2006–2012, while the rate of employment in the formal private sector did not change. Wahba (2009) studies the labor mobility between formal and informal sectors and finds that the highly educated individuals are likely to move from the informal to the formal sector, while the less educated individuals remain in the informal sector. Elsayed and Wahba (2018) investigate the informal sector and labor mobility after the Arab Spring revolution in Egypt. They demonstrate that informal employment increased significantly after the revolution due to the stagnation of the highly educated individuals in the informal sector, and less educated individuals were likely to lose their formal status and become informally employed. These studies reveal that the informal sector in Egypt represents a major part of the labor market, while the formal private sector has ceased growing.

Previous literature has mainly studied the wage gap between two sectors (e.g., public–private or formal–informal), whereas the labor market in developing countries consists of three sectors (e.g., public, formal private and informal private sectors). Additionally, the previous studies do not focus on addressing the relationship among the three sectors over time in developing countries in which the public sector is the main source of employment, and there are very large informal private and small formal private sectors. Thus, this paper studies the wage gap, labor mobility and the impact of changing jobs across the public, formal private and informal private sectors over time within a single framework. A novelty in this research lies in including the three sectors in an analytic framework with panel data at individual level and focusing on examining changes in wages and labor over time. This approach enables us to identify sources of the temporal wage gap and labor mobility among the three sectors as well as individuals who stay in one sector or move to another sector over time with an associated wage impact. To this end, we use Oaxaca–Blinder decomposition and difference-in-difference methods with the Egyptian Labor Market Panel Survey (ELMPS).

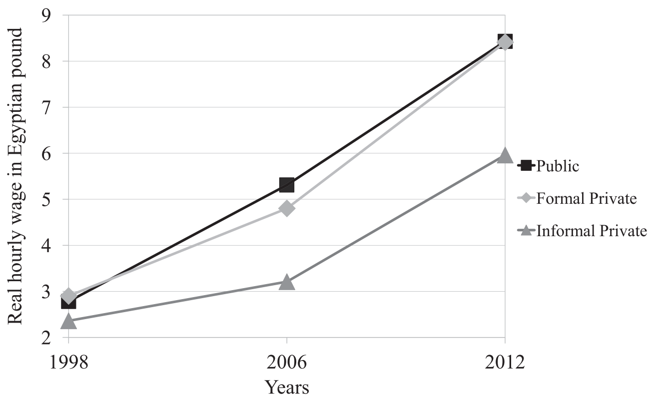

Fig. 1. Hourly wage per sector.

2. Methods and materials

2.1. Summary of data

This paper uses data from the Egyptian Labor Market Panel Survey (ELMPS) (ELMPS, 1998, 2006, 2012; Assaad and Krafft, 2013). ELMPS is a panel data set that consists of three waves, 1998, 2006 and 2012, which enables us to investigate (i) the factors that explain the wage gap between public and private sectors over time, (ii) labor mobility and (iii) the impact of changing employment sectors on wages. The definition of the labor force used in this paper is "all the individuals between the age 15 and 65 including students because some of them work after school." However, those unemployed or self-employed (including agriculture workers) in the three waves are excluded because the unemployed individuals do not affect the wage gap between the two sectors, while most of the self-employed workers do not report their hourly wage in ELMPS. We distinguish between the formal private sector and the informal private sector. Using the definition of the formal employment by Assaad and Krafft (2015), "formal jobs are the jobs that have a contract and/or social insurance coverage, while the informal jobs are those with neither a contract nor social insurance". The difference in real hourly wages for the public sector, formal private sector and informal private sector is shown in Fig. 1. In 1998, the wage gap between the three sectors was very narrow; however, this gap expanded in 2006. The wage gap between public and formal private sectors becomes almost 0.5 LE or Egyptian pound,[1] while this gap was nullified in 2012. The informal sector workers had a wage penalty during the three waves and this penalty increased over time.

Fig. 2 shows that more than half (54%) of the observations represented people who were working in the public sector in 1998, while this decreased to 44% in 2006 and 41% in 2012. On the other hand, the percentage of workers in the formal private sector increased from 16% in 1998 to 19% in 2006, then decreased to 16% in 2012 due to the instability following the Arab Spring revolution in 2011. The informal-private sector was the only sector that expanded in employment over time by increasing from 30% in 1998 to almost 43% in 2012. Figs. 1 and 2 show that even though the informal sector average wage was lower than the average wage in the other two sectors, the percentage of employed individuals in that sector increased over time, which indicates that there was a high proportion of labor in Egypt suffering from poor work conditions (e.g., lower wages, low insurance coverage and instability).

2.2. Analytical framework

Our analysis to determine the wage gap among the public (PUB), formal private (FP) and informal private (IP) sectors is based on three different earning equations. We consider the following three wage equations:

$$ln(W_i^{PUB}) = \beta^{PUB}X_i + \epsilon_i^{PUB} \tag{1}$$

$$ln(W_i^{FP}) = \beta^{FP}X_i + \epsilon_i^{FP} \tag{2}$$

$$ln(W_i^{IP}) = \beta^{IP}X_i + \epsilon_i^{IP} \tag{3}$$

where W_i^{PUB}, W_i^{FP} and W_i^{IP} are, respectively, the hourly wages for the public, formal private, and informal private sectors for individual i; X is a vector of the independent variables; β^{PUB}, β^{FP} and β^{IP} are, respectively, the coefficients or the returns of variables; and ϵ_i^{PUB}, ϵ_i^{FP} and ϵ_i^{IP} are the error terms of each equation.

There are three sectors in our sample (the public, formal private, and informal private sectors). To identify wage differentials among the three sectors, we consider two-subsample analysis and apply the decomposition method,

[1] The LE represents the national currency, the Egyptian pound.

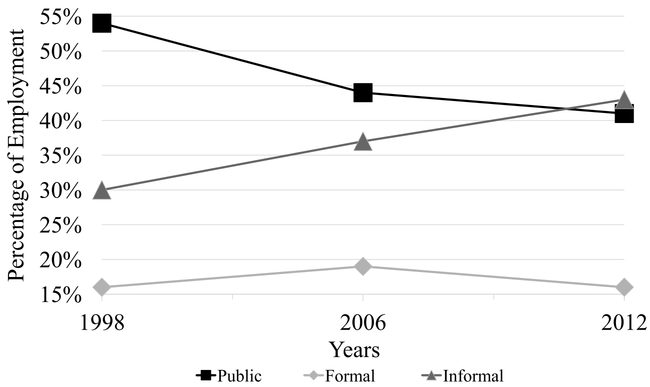

Fig. 2. Percentage of employment per sector.

following Oaxaca (1973), Blinder (1973), Reimers (1983), Neumark (1988) and Oaxaca and Ransom (1994) for each subsample. The first subsample consists of workers in the public and formal private sectors, dropping workers in the informal private sector. The second subsample consists of workers in the public and informal private sectors, dropping workers in the formal private sector. For each subsample, the decomposition is obtained by estimating the following equation ($K \in \{FP, IP\}$):

$$\overline{ln(W^K)} - \overline{ln(W^{PUB})} = \underbrace{\hat{\beta}^{\star}(\bar{X}^K - \bar{X}^{PUB})}_{\text{Explained Effect}} + \underbrace{(\hat{\beta}^K - \hat{\beta}^{\star})\bar{X}^K + (\hat{\beta}^{\star} - \hat{\beta}^{PUB})\bar{X}^{PUB}}_{\text{Unexplained Effect}} \tag{4}$$

where $\bar{X}^K$ and $\bar{X}^{PUB}$ are, respectively, the mean of the independent variables for formal (informal) private and public sectors; $\hat{\beta}^K$ and $\hat{\beta}^{PUB}$ are, respectively, the ordinary least squares (OLS) estimates of coefficients of $\hat{\beta}^K$ and $\hat{\beta}^{PUB}$ in the wage equations in the formal (informal) private and public sectors. $\hat{\beta}^{\star}$ is a vector of estimates of coefficients obtained from the pooled regression of two sectors (i.e., public–formal private sectors in the first sub-sample and public–informal private sectors in the second sub-sample). The first part of Eq. (4) quantifies the wage gap that derives from characteristics, calculating the difference in the means of the regressors between the formal (informal) private and public sectors weighted by the estimated coefficients of the pooled regressions in public–formal (public–informal) sectors. It represents the component due to the differences in employees' characteristics, such as education, work experience and so on, being interpreted to be "explained effect".

The second part in Eq. (4) quantifies the wage gap that derives from sectoral differences. It calculates the difference between the estimated coefficients of the regression in the formal (informal) private sector and those of the pooled regression in public–formal (public–informal) private sectors weighted by the means of the regressors, $\bar{X}^K, K \in \{FP, IP\}$, representing the component due to the advantage and disadvantage of the formal (informal) private sector as compared with the public sector. Likewise, the third part in Eq. (4) quantifies the wage gap that derives from sectoral differences, calculating the difference between the estimated coefficients of the regression in the public sector and those of the pooled regression in public–formal (public–informal) private sectors weighted by the means of the regressors, $\bar{X}^{PUB}$, representing the component due to the advantage and disadvantage of the public sector as compared with the formal (informal) private sector. The last two parts in Eq. (4) are interpreted to represent the difference in the structure of wages between two sectors as "unexplained effect".

We next evaluate the wage differentials among the three sectors by applying the difference-in-difference method over each of six subsamples consisting of the workers who stayed in the same sector for two waves and those who changed the sector that they work in between two waves, which correspond to the control and treatment groups, respectively (Pages and Stampini, 2009; Elsayed and Wahba, 2018). Specifically, we consider six subsample analyses: (i) subsample 1 consists of the workers who stayed in the public sector and those who moved from that sector to the formal private sector; (ii) subsample 2 consists of the workers who stayed in the public sector and those who moved from that sector to the informal private sector; (iii) subsample 3 consists of the workers who stayed in the formal private sector and those who moved from that sector to the public sector; (iv) subsample 4 consists of the workers who stayed in the formal private sector and those who moved from that sector to the informal private sector; (v) subsample 5 consists of the workers who stayed in the informal private sector and those who moved from that sector to the public sector; and (vi) subsample 6 consists of the workers who stayed in the informal private sector and those who moved from that sector to the formal private sector.

Our panel data consist of three waves (e.g., 1998, 2006, and 2012). We conduct empirical analysis over the first two waves between 1998 and 2006 and the last two waves between 2006 and 2012. The estimated models are described by

$$ln(W_{it}) = \alpha + \tau T_{it} + \gamma R_{it} + \delta T_{it} R_{it} + \epsilon_{it}, \tag{5}$$

Table 1
Summary statistics for 1998.

	Public sector			Formal private sector			Informal private sector			Total		
	Mean	p(50)	SD	Mean	p(50)	SD	Mean	p(50)	SD	Mean	p(50)	SD
Hourly wage	2.78	2.38	1.44	2.9	2.5	1.66	2.36	2.11	1.54	2.68	2.31	1.51
Male	0.69	–	–	0.91	–	–	0.87	–	–	0.78	–	–
Married	0.8	–	–	0.7	–	–	0.42	–	–	0.67	–	–
Age	40.24	40	9.99	37.41	36	11.27	29.27	26	11.57	36.49	36	11.74
Work experience	19.49	19	11.7	19.04	18	13.36	13.36	9	12.53	17.58	16	12.54
Part time job	0	–	–	0.01	–	–	0.08	–	–	0.03	–	–
Education												
Illiterate	0.05	–	–	0.1	–	–	0.28	–	–	0.13	–	–
Read and write	0.06	–	–	0.14	–	–	0.12	–	–	0.09	–	–
Primary school	0.11	–	–	0.21	–	–	0.27	–	–	0.18	–	–
Preparatory school	0.32	–	–	0.24	–	–	0.24	–	–	0.28	–	–
Secondary school	0.13	–	–	0.07	–	–	0.03	–	–	0.09	–	–
University	0.3	–	–	0.24	–	–	0.05	–	–	0.21	–	–
Postgraduate	0.02	–	–	0.01	–	–	0	–	–	0.01	–	–
Occupation												
Management	0.09	–	–	0.27	–	–	0.08	–	–	0.12	–	–
Profession	0.37	–	–	0.14	–	–	0.03	–	–	0.23	–	–
Technician	0.12	–	–	0.04	–	–	0.01	–	–	0.07	–	–
Clerk	0.17	–	–	0.06	–	–	0.01	–	–	0.11	–	–
Worker	0.12	–	–	0.13	–	–	0.24	–	–	0.16	–	–
Craftsman	0.08	–	–	0.17	–	–	0.49	–	–	0.22	–	–
Operator	0.04	–	–	0.17	–	–	0.05	–	–	0.07	–	–
Elementary	0.01	–	–	0.01	–	–	0.08	–	–	0.03	–	–
Residence area												
Greater Cairo area	0.28	–	–	0.44	–	–	0.32	–	–	0.32	–	–
Alexandria and Canal area	0.17	–	–	0.19	–	–	0.14	–	–	0.17	–	–
Urban Upper Egypt area	0.22	–	–	0.13	–	–	0.15	–	–	0.18	–	–
Rural Upper Egypt area	0.06	–	–	0.02	–	–	0.1	–	–	0.07	–	–
Urban Lower Egypt area	0.15	–	–	0.14	–	–	0.16	–	–	0.15	–	–
Rural Lower Egypt area	0.09	–	–	0.06	–	–	0.09	–	–	0.09	–	–
Industry												
Manufacturing industry	0.12	–	–	0.28	–	–	0.25	–	–	0.19	–	–
Electricity industry	0.03	–	–	0	–	–	0	–	–	0.01	–	–
Construction industry	0.02	–	–	0.05	–	–	0.19	–	–	0.08	–	–
Trade industry	0.02	–	–	0.34	–	–	0.36	–	–	0.19	–	–
Transportation industry	0.07	–	–	0.16	–	–	0.05	–	–	0.08	–	–
Financial industry	0.03	–	–	0.05	–	–	0.02	–	–	0.03	–	–
Service industry	0.70	–	–	0.11	–	–	0.12	–	–	0.40	–	–
Mining industry	0	–	–	0.01	–	–	0	–	–	0	–	–
% of employment		54%			16%			30%			100%	

where W_{it} is the hourly wage for individual i at time t, T_{it} is a dummy variable for the treatment, which takes a value of one when a worker changes her sector from (i) the public sector to the formal private sector; (ii) the public sector to the informal private sector; (iii) the formal private sector to the public sector; (iv) the formal private sector to the informal private sector; (v) the informal private sector to the public sector; and (vi) the informal private sector to the formal private sector. R_{it} is a dummy variable for time period, which equals zero for the base period and one for the second period. The base year is 1998 for the analysis over the first two waves and 2006 for the analysis over the last two waves.

3. Results

Tables 1–3 present the summary statistics for all of the variables in the three years of 1998, 2006 and 2012, respectively. In 1998, the average wage in the formal private sector is higher than those in the public and informal private sectors, while the public sector pays higher wages than the informal private sector. Tables 2 and 3 show that the wage in the public sector is the highest among the three sectors and that there is a wage penalty for working in the informal private sector compared to the formal private sector in 2006 and 2012. In summary, Tables 1–3 show an overall trend of the wage gap among the three sectors. The wage gap between the public and formal private sectors is in favor of the private sector in 1998, while this gap is in favor of the public sector in 2006 and 2012. The informal private sector consistently has a wage penalty, compared to the other two sectors, and this penalty increases over the years.

The average age in the overall sample for 1998, 2006 and 2012 is approximately 35 years old. Employees in the public sector are, on average, 5 years older than their counterparts in the formal private sector and their working experience is approximately three years longer than their counterpart in the formal private sector. The informal private sector has the youngest employees, with an average age of 30 years old, which is consistent with the fact that the informal

Table 2
Summary statistics for 2006.

	Public sector			Formal private sector			Informal private sector			Total		
	Mean	p(50)	SD	Mean	p(50)	SD	Mean	p(50)	SD	Mean	p(50)	SD
Hourly wage	5.31	3.53	12.12	4.8	3.4	3.4	3.21	2.67	3.83	4.53	3.2	9.41
Male	0.69	–	–	0.88	–	–	0.88	–	–	0.8	–	–
Married	0.85	–	–	0.72	–	–	0.51	–	–	0.7	–	–
Age	40.76	41	10.12	35.41	33	10.89	29.81	27	10.69	35.67	34	11.58
Work experience	20.52	20	11.54	16.93	14	12.39	13.89	11	11.44	17.37	15	12.04
Part time job	0	–	–	0.01	–	–	0.04	–	–	0.02	–	–
Education												
Illiterate	0.06	–	–	0.09	–	–	0.23	–	–	0.13	–	–
Informal private sector	0.04	–	–	0.08	–	–	0.08	–	–	0.06	–	–
Primary school	0.09	–	–	0.16	–	–	0.23	–	–	0.16	–	–
Preparatory school	0.37	–	–	0.34	–	–	0.35	–	–	0.36	–	–
Secondary school	0.08	–	–	0.06	–	–	0.03	–	–	0.06	–	–
University	0.35	–	–	0.27	–	–	0.08	–	–	0.23	–	–
Postgraduate	0.02	–	–	0.01	–	–	0	–	–	0.01	–	–
Occupation												
Management level	0.09	–	–	0.15	–	–	0.07	–	–	0.1	–	–
Profession	0.35	–	–	0.17	–	–	0.03	–	–	0.2	–	–
Technician	0.22	–	–	0.09	–	–	0.04	–	–	0.13	–	–
Clerk	0.1	–	–	0.04	–	–	0.02	–	–	0.06	–	–
Worker	0.13	–	–	0.15	–	–	0.25	–	–	0.18	–	–
Craftsman	0.05	–	–	0.17	–	–	0.45	–	–	0.22	–	–
Operator	0.04	–	–	0.19	–	–	0.09	–	–	0.09	–	–
Elementary	0.01	–	–	0.02	–	–	0.06	–	–	0.03	–	–
Residence area												
Greater Cairo area	0.24	–	–	0.33	–	–	0.26	–	–	0.26	–	–
Alexandria and Canal area	0.16	–	–	0.21	–	–	0.13	–	–	0.16	–	–
Urban Upper Egypt area	0.2	–	–	0.13	–	–	0.13	–	–	0.16	–	–
Rural Upper Egypt area	0.1	–	–	0.07	–	–	0.16	–	–	0.12	–	–
Urban Lower Egypt area	0.13	–	–	0.13	–	–	0.13	–	–	0.13	–	–
Rural Lower Egypt area	0.14	–	–	0.11	–	–	0.14	–	–	0.13	–	–
Industry												
Manufacturing industry	0.08	–	–	0.28	–	–	0.22	–	–	0.18	–	–
Electricity industry	0.03	–	–	0.01	–	–	0	–	–	0.01	–	–
Construction industry	0.02	–	–	0.05	–	–	0.21	–	–	0.1	–	–
Trade industry	0.02	–	–	0.26	–	–	0.35	–	–	0.2	–	–
Transportation industry	0.06	–	–	0.18	–	–	0.09	–	–	0.09	–	–
Financial industry	0.03	–	–	0.02	–	–	0	–	–	0.01	–	–
Service industry	0.75	–	–	0.2	–	–	0.14	–	–	0.39	–	–
Mining industry	0	–	–	0.01	–	–	0	–	–	0	–	–
% of employment		44%			19%			37%			100%	

private sector has the least experienced individuals, with an average of 13 working years. The occupation variables are defined as management, profession, technician, clerk, worker, craftsman, operator, elementary categories and so on, following Central agency for public mobilization and statistics (1985). Approximately 50% of employees in the public sector work as technicians and professions in the three waves, while 30% of workers in the formal private sector serve as managers or professions. On average, 46% of the informal private sector employees are craftsmen. Employees in different sectors are concentrated in specific areas of Egypt. More than 50% of employees in the formal and informal private sector employees are concentrated in urban and big cities (e.g., Greater Cairo and Alexandria), while the public sector employees are equally distributed across the areas of residence.

Table 1 shows that 77% of employees in the public sector have a high level of education (e.g., preparatory or higher) in 1998, while the percentages in the formal and informal private sectors are 56% and 32%, respectively. Table 2 presents the percentage of employment with higher education in 2006 and shows that the education level in the public, formal and informal sectors increases to become 82%, 68% and 46%, respectively, which confirms that highly educated employees in the public sector represent 86% of public overall employment in 2012, and the percentage of highly educated employees in the formal and informal private sectors are 76% and 52%, respectively. These results show that the public sector always has a higher percentage of highly educated employees than the other two sectors, while a considerable portion of highly educated people end up working in the informal sector.

Tables 1–3 present the different industries in each employment sector and the percentage of employment in each of these industries. The service industry represents the main employer in the public sector with 70% or higher over the period from 1998 to 2012. Trade and manufacturing industries represent the main employment industries for the formal (informal) private sector with around 34% and 28% (36% and 25%) in 1998 and around 26% and 28% (35% and 22%) in 2006, respectively. Manufacturing and service industries employ 28% and 26% in 2012, respectively. On the other hand, trade and construction industries are the main employers in the informal private sector in 2012 with 31% and 27%, respectively.

Table 3
Summary statistics for 2012.

	Public sector			Formal private sector			Informal private sector			Total		
	Mean	p(50)	SD	Mean	p(50)	SD	Mean	p(50)	SD	Mean	p(50)	SD
Hourly wage	8.43	6.48	9.58	8.41	5.62	11.26	5.96	5	6.05	7.45	5.7	8.77
Male	0.67	–	–	0.9	–	–	0.93	–	–	0.82	–	–
Married	0.88	–	–	0.76	–	–	0.63	–	–	0.75	–	–
Age	40.8	41	10.34	35.48	33	10.32	31.5	29	10.1	35.92	34	11.08
Work experience	19.04	18	11.31	14.91	12	10.83	13.86	12	10.34	16.15	14	11.09
Part time job	0.01	–	–	0.02	–	–	0.13	–	–	0.07	–	–
Education												
Illiterate	0.04	–	–	0.07	–	–	0.2	–	–	0.11	–	–
Read and write	0.03	–	–	0.04	–	–	0.05	–	–	0.04	–	–
Primary school	0.08	–	–	0.13	–	–	0.23	–	–	0.15	–	–
Preparatory school	0.38	–	–	0.36	–	–	0.39	–	–	0.38	–	–
Secondary school	0.07	–	–	0.05	–	–	0.03	–	–	0.05	–	–
University	0.38	–	–	0.34	–	–	0.1	–	–	0.25	–	–
Postgraduate	0.03	–	–	0.01	–	–	0	–	–	0.02	–	–
Occupation												
Management level	0.07	–	–	0.13	–	–	0.07	–	–	0.08	–	–
Profession	0.39	–	–	0.2	–	–	0.03	–	–	0.21	–	–
Technician	0.24	–	–	0.1	–	–	0.03	–	–	0.13	–	–
Clerk	0.07	–	–	0.04	–	–	0.01	–	–	0.04	–	–
Worker	0.06	–	–	0.12	–	–	0.21	–	–	0.13	–	–
Craftsman	0.03	–	–	0.1	–	–	0.45	–	–	0.22	–	–
Operator	0.04	–	–	0.26	–	–	0.13	–	–	0.12	–	–
Elementary	0.1	–	–	0.07	–	–	0.07	–	–	0.08	–	–
Residence area												
Greater Cairo area	0.19	–	–	0.36	–	–	0.18	–	–	0.21	–	–
Alexandria and Canal area	0.15	–	–	0.19	–	–	0.11	–	–	0.14	–	–
Urban Upper Egypt area	0.18	–	–	0.1	–	–	0.14	–	–	0.15	–	–
Rural Upper Egypt area	0.15	–	–	0.07	–	–	0.23	–	–	0.17	–	–
Urban Lower Egypt area	0.12	–	–	0.11	–	–	0.11	–	–	0.11	–	–
Rural Lower Egypt area	0.19	–	–	0.15	–	–	0.19	–	–	0.18	–	–
Industry												
Manufacturing industry	0.07	–	–	0.28	–	–	0.18	–	–	0.15	–	–
Electricity industry	0.05	–	–	0.01	–	–	0	–	–	0.02	–	–
Construction industry	0.02	–	–	0.05	–	–	0.27	–	–	0.14	–	–
Trade industry	0.01	–	–	0.2	–	–	0.31	–	–	0.18	–	–
Transportation industry	0.05	–	–	0.17	–	–	0.1	–	–	0.09	–	–
Financial industry	0.02	–	–	0.02	–	–	0	–	–	0.01	–	–
Service industry	0.78	–	–	0.26	–	–	0.15	–	–	0.4	–	–
Mining industry	0	–	–	0	–	–	0	–	–	0	–	–
% of employment		41%			16%			43%			100%	

The Oaxaca–Blinder decomposition results of wage gaps over time for the formal private–public and informal private–public sectors are presented in Tables 4 and 5, respectively. Tables 4 and 5 show the total log wage difference and the decompositions into two parts: explained and unexplained. Regarding Table 4, the sign of the "total explained" part in 1998 is negative, which is different from the sign of the total log wage gap (e.g., positive), which means that the sum of the differences in observable characteristics (total explained) does not contribute to the total log wage gap. Therefore, an explained variable with a negative (positive) sign is interpreted to decrease (increase) the private formal–public wage gap because the total log wage difference in 1998 is positive. In 2006 and 2012, the sign of the "total explained" part is the same as the sign of the total log wage gap (e.g., negative), implying that the differences in observable characteristics contribute to the total wage gap. The explained variable with a negative (positive) sign is interpreted to increase (decrease) the wage gap because the total log wage gap in 2006 and 2012 are negative. These interpretations mean that the observable variables with negative signs in explained parts do not contribute to the total raw wage gap in 1998 but do contribute to the wage gap in 2006 and 2012.

Table 4 demonstrates the decomposition results between formal private–public sectors. The "total explained" part contributes to the "total log wage gap" in 2006 and 2012 but does not contribute in 1998. The age, work experience and education variables are statistically significant at 1%, the married variable is statistically significant at 10%, and they contribute to the "total explained" part of wage gap in 1998 by 45% ($= \frac{-0.049}{-0.108} \times 100$), 26%, 48% and 4.6 %, respectively. In 2006 (2012), the married, age, work experience, education and occupation variables also contribute to the "total explained" part by 5.6 %, 64%, 17%, 22% and 22% (7%, 67%, 20%, 28% and 28%), respectively. On the other hand, the male (residence area) variable reduces the "total explained" part by 5.5 %, 17% and 25% (18%, 17% and 27%) at 1% statistical significance in 1998, 2006 and 2012, respectively. These results show that the total log wage gap between the formal private and public sector originates from the differences in employees' age, education and work experience in 2006 and

Table 4
Oaxaca–Blinder for formal private–public.

Variables	(1) 1998	(2) 2006	(3) 2012
Explained			
Married	−0.005* (0.003)	−0.008** (0.004)	−0.008*** (0.003)
Male	0.006** (0.003)	0.024*** (0.004)	0.028*** (0.005)
Age	−0.049*** (0.009)	−0.090*** (0.016)	−0.076*** (0.009)
Work experience	−0.028*** (0.006)	−0.025** (0.012)	−0.023*** (0.009)
Part time job	0.003 (0.002)	0.003 (0.002)	0.002 (0.002)
Education[a]	−0.052*** (0.007)	−0.032*** (0.006)	−0.032*** (0.006)
Occupation[b]	−0.002 (0.010)	−0.032*** (0.011)	−0.033*** (0.011)
Residence area[c]	0.019*** (0.003)	0.019*** (0.004)	0.030*** (0.004)
Total explained	−0.108*** (0.014)	−0.141*** (0.015)	−0.113*** (0.015)
Total unexplained	0.135*** (0.019)	0.092*** (0.022)	0.044*** (0.020)
Total log wage gap	0.027 (0.020)	−0.050** (0.020)	−0.069*** (0.020)
Observations	3088	4516	5307

Robust standard errors in parentheses.
***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.
[a]Education is taken to be the sum of the dummy variables for education, taking illiterate as the base group in Oaxaca–Blinder decomposition. This variable includes read and write, primary school, preparatory school, secondary school, university and postgraduate.
[b]Occupation is taken to be the sum of the dummy variables for occupation, taking elementary jobs as the base group. This variable includes management, profession, technician, clerk, worker, craftsman and operator in Oaxaca–Blinder decomposition.
[c]Residence area is taken to be the sum of the dummy variables for area dummy, taking Greater Cairo area as the base group. This variable includes Alexandria and Canal area, Urban Upper, Rural Upper, Urban Lower and Rural Lower in Oaxaca–Blinder decomposition.

2012 and also show the fact that employees in the public sector have higher age and education level on average than employees in the formal private sector and remain long term in the same sector once they obtain a position there.

The decomposition results for the informal private–public wage gaps over years are presented in Table 5. The "total explained" part contributes to the "total log wage gap" for all three years. The married, age, work experience and education variables are statistically significant and contribute to the "total explained" part by 4.6 %, 36%, 17% and 47% in 1998; 7.5 %, 41%, 16% and 34% in 2006; and 9.3 %, 49%, 8% and 53% in 2012, respectively.[2] The male and part time job variables reduce the "total explained" part of the wage gap, and their magnitudes are considered small over the period from 1998 to 2012. The results indicate that the total log wage gap between the two sectors is mainly explained by differences in employees' average age, work experience and educational level between the informal private sector and public sector.

Tables 4 and 5 reveal that the age, work experience and education variables contribute to the explained part of the total log wage gap between the formal (informal) private and public sectors on average over the three waves by 57%, 21% and 33% (42%, 14% and 39%), respectively.[3] In summary, these three variables are the main contributors to the total log wage gap over the period from 1998 to 2012 between the formal (informal) private and public sectors. The results in Tables 4 and 5 imply that the wage gap between the public and the formal (informal) private sectors remains strong; therefore, the public sector attracts and retains the highly educated and experienced employees. On the other hand, the informal private sector employment witnessed a rapid growth in the employment share of the labor market from 1998 to 2012 (see Fig. 2), which may have hindered the development of the formal private sector. These results are consistent

2 The occupation variable contributes only in 2006 by 16% of the "total explained" part at 5% statistical significance.

3 The three percentages associated with age, work experience and education in Tables 4 and 5 are calculated, respectively, as follows:

$$= \left[\left(\frac{\text{Variable coefficient 1998}}{\text{Total explained 1998}} + \frac{\text{Variable coefficient 2006}}{\text{Total explained 2006}} + \frac{\text{Variable coefficient 2012}}{\text{Total explained 2012}} \right) \div 3 \right] \times 100.$$

.

Table 5
Oaxaca–Blinder for informal private–public.

Variables	(1) 1998	(2) 2006	(3) 2012
Explained			
Married	−0.013**	−0.020***	−0.018***
	(0.005)	(0.007)	(0.005)
Male	0.007**	0.027***	0.036***
	(0.003)	(0.004)	(0.005)
Age	−0.100***	−0.108***	−0.095***
	(0.014)	(0.022)	(0.012)
Work experience	−0.046***	−0.042***	−0.016**
	(0.009)	(0.014)	(0.007)
Part time job	0.015***	0.009***	0.023***
	(0.004)	(0.002)	(0.003)
Education[a]	−0.131***	−0.090***	−0.103***
	(0.009)	(0.009)	(0.012)
Occupation[b]	−0.010	−0.042**	−0.021
	(0.019)	(0.019)	(0.023)
Residence area[c]	0.001	0.001	0.001
	(0.002)	(0.002)	(0.002)
Total explained	−0.277***	−0.264***	−0.193***
	(0.018)	(0.018)	(0.016)
Total unexplained	0.118***	−0.068***	−0.093***
	(0.018)	(0.024)	(0.020)
Total log wage gap	−0.159***	−0.332***	−0.285***
	(0.011)	(0.013)	(0.012)
Observations	3670	5648	7373

Robust standard errors in parentheses.
***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.
[a]Education is taken to be the sum of the dummy variables for education, taking illiterate as the base group in Oaxaca–Blinder decomposition. This variable includes read and write, primary school, preparatory school, secondary school, university and postgraduate.
[b]Occupation is taken to be the sum of the dummy variables for occupation, taking elementary jobs as the base group. This variable includes management, profession, technician, clerk, worker, craftsman and operator in Oaxaca–Blinder decomposition.
[c]Residence area is taken to be the sum of the dummy variables for area dummy, taking Greater Cairo area as the base group. This variable includes Alexandria and Canal area, Urban Upper, Rural Upper, Urban Lower and Rural Lower in Oaxaca–Blinder decomposition.

with Assaad (1997) in that the employees with higher education in Egypt are usually attracted to the public sector and the return to education was higher in the public sector than in the formal private sector in 1988.

We apply difference-in-difference (DID) method to clarify the labor mobility over time among the different sectors as well as the impacts of changing employment sectors on wages, and Tables 6 and 7 show the results of DID estimations over the periods of 1998–2006 and 2006–2012, respectively.[4] First, we report the labor mobility among the three sectors and next discuss the impacts of changing employment sectors on wages. Tables 6 and 7 show that, from 1998 to 2006 (from 2006 to 2012), 24% (18%) of workers are identified to be movers from the informal private sector to the formal private sector, while 20% (35%) of workers are movers from the formal private sector to the informal private sector, implying that the net labor mobility to the formal private sector from the informal private one has been negligible or even negative. Regarding the labor mobility to the public sector from the formal (informal) private sector, we can see that the percentage of movers from the formal (informal) private sector to the public sector, e.g., 27% (14%) in 2006–2012, is more dominant than the counterpart from the public sector to the formal (informal) private sector, e.g., 3% (2%) in 2006–2012. This implies that the public sector has been attracting labor from both formal and informal private sectors.

Tables 6 and 7 also show the impact of changing employment sectors on wages by comparing movers from one sector to another sector with stayers between two periods from 1998 to 2006 and from 2006 to 2012. Movers from the formal private sector to the informal private sector are demonstrated to incur wage losses of 19.2 (20.8) log points between 1998 and 2006 (2006 and 2012) at 5% (1%) significance level. Movers from the public sector to the informal private sector are identified to have wage losses by 21.2 (2.6) log points at 10% significance level (insignificant level) between 1998 and 2006 (2006 and 2012). Overall, it can be concluded that movers from the formal private (or public) sector to the informal private sector tend to experience significant (less significant) wage losses. On the other hand, movers from the informal private to the formal private sectors between 1998 and 2006 (2006 and 2012) are identified to have a wage gain by 12.4

[4] We include the results of DID methods for some other independent variables in the appendices.

Table 6
Difference-in-difference in log hourly wages between 1998 and 2006.

	(1) Informal[a] to formal[b]	(2) Formal to informal	(3) Informal to public	(4) Public to informal	(5) Formal to public	(6) Public to formal
	1998 ($R = 0$)					
Remained in sector (C)	0.437	0.347	0.468	−0.102	0.357	−0.111
	(0.084)	(0.161)	(0.087)	(0.091)	(0.181)	(0.086)
Moved to another (T)	0.467	0.342	0.426	−0.191	0.295	−0.060
	(0.093)	(0.170)	(0.105)	(0.105)	(0.199)	(0.089)
Diff ($T - C$)	0.0299	−0.005	−0.042	−0.089*	−0.061	0.051
	(0.037)	(0.056)	(0.045)	(0.053)	(0.080)	(0.043)
	2006 ($R = 1$)					
Remained in sector (C)	0.827	0.876	0.859	0.482	0.886	0.470
	(0.083)	(0.177)	(0.086)	(0.092)	(0.195)	(0.087)
Moved to another (T)	0.981	0.679	0.791	0.181	0.698	0.641
	(0.105)	(0.172)	(0.127)	(0.153)	(0.220)	(0.132)
Diff ($T - C$)	0.154**	−0.197***	−0.067	−0.301**	−0.188	0.167
	(0.068)	(0.074)	(0.078)	(0.121)	(0.126)	(0.113)
Diff-in-Diff	0.124*	−0.192**	−0.025	−0.212*	−0.126	0.116
	(0.067)	(0.085)	(0.087)	0.125	0.130	(0.109)
No. of obs. controlled group	578	342	578	2964	342	2964
No. of obs. treated group	186	68	102	42	70	82
Total no. of observations	764	410	680	3006	412	3046
% of individuals that changed the sector[c]	24%	20%	15%	1%	17%	3%

Robust Standard Errors in Parentheses.
***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.
[a]"Informal" refers to informal private sector.
[b]"Formal" refers to formal private sector.
[c]% of individuals that changed the sector $= \frac{\text{No. of obs. treated group}}{\text{Total no. of observation}}$.

(19.6) log points at 10% (1%) statistical significance. These results indicate that changing the employment sectors does not necessarily affect the wages except for the cases in which people move to the informal private sector or move from the informal private sector to the formal private sector. Wage loss (gain) is expected for movers from the formal (informal) private sectors to the informal (formal) private sectors, while movers to and/or from the public sectors do not experience any wage change.

The DID results imply that workers in the formal private sector are more unstable and are likely to face a high risk of falling into the informal private sector as compared with workers in the public sector. The percentages and wage losses of movers from the public sector to the informal private sector are consistently much smaller than the percentages and wage losses of movers from the formal private sector to the informal private sector over the periods of 1998–2006 and 2006–2012, which implies that a majority of movers to the informal private sector come from the formal private sector in Egypt and suffer from wage losses. These results can be considered important empirical evidence of how the public sector has been more attractive than any other sector in Egypt and how employment in the informal (formal) private sector has expanded (shrunk).

Overall, the Oaxaca–Blinder decomposition and DID methods demonstrate that highly educated people are attracted only to the public sector and stay there in the long term. The paper also reveals that Egyptian employees in the formal private sector face a high risk of unwillingly falling into and staying in the informal private sector, these results can be considered to be in line with previous literature. Fernández-de-Córdoba et al. (2012) and Francisco de Castro and Matteo Salto (2013) study the wage gap between public and formal private sectors in some EU and OECD countries and find that public sector workers have wage premium compared to the private sector workers. This gap appears more between highly educated and experienced workers in the two sectors. The public sector in Egypt and most of developing countries has a wage premium for highly educated and experienced workers. However, the existence of a growing informal sector in developing countries represents the main challenge for the labor market in these countries to achieve social welfare. The informal sector has poor work conditions for the workers (e.g., lower wages, low insurance coverage and instability) and negative consequences for the economy (e.g., high tax evasion and low productivity).

Studying wage gap and labor mobility between public and formal private sectors only will give misleading findings about the labor markets because the informal sector has a significant share of employment in developing countries. Our paper provides a comprehensive picture about the labor market in Egypt by studying the three sectors in a single analytical framework. Thus, the results of our paper can be considered likely obstacles for further economic growth and the stability of the Egyptian economy. The Egyptian government should restructure the wage system and other benefits associated with employment, considering a balance with private sectors and providing people with incentive schemes to nurture (formalize) the formal (informal) private sector. However, it is reported that Egyptians are more risk averse, preferring

Table 7
Difference-in-difference in log hourly wages between 2006 and 2012.

	(1) Informal[a] to formal[b]	(2) Formal to informal	(3) Informal to public	(4) Public to informal	(5) Formal to public	(6) Public to formal
	2006 ($R = 0$)					
Remained in sector (C)	0.861	0.722	0.871	0.391	0.711	0.373
	(0.080)	(0.137)	(0.091)	(0.125)	(0.177)	(0.124)
Moved to another (T)	0.796	0.714	0.914	0.313	0.666	0.472
	(0.083)	(0.131)	(0.101)	(0.128)	(0.179)	(0.135)
Diff ($T - C$)	−0.065**	−0.008	0.0431	−0.077	−0.045	0.0998
	(0.030)	(0.052)	(0.056)	0.056	(0.063)	(0.060)
	2012 ($R = 1$)					
Remained in sector (C)	1.483	1.353	1.494	1.099	1.341	1.081
	(0.081)	(0.135)	(0.091)	(0.124)	(0.179)	(0.124)
Moved to another (T)	1.614	1.136	1.469	0.996	1.316	1.117
	(0.091)	(0.131)	(0.108)	(0.164)	(0.194)	(0.144)
Diff ($T - C$)	0.131***	−0.216***	−0.025	−0.104	−0.025	0.036
	(0.046)	(0.057)	(0.060)	0.097	(0.085)	(0.081)
Diff-in-Diff	0.196***	−0.208***	−0.068	−0.026	0.020	−0.064
	(0.052)	(0.070)	(0.076)	(0.104)	(0.089)	(0.074)
No. of obs. controlled group	1470	564	1470	4024	564	4024
No. of obs. treated group	320	306	234	84	210	130
Total no. of observations	1790	870	1704	4108	774	4154
% of individuals that changed the sector[c]	18%	35%	14%	2%	27%	3%

Robust Standard Errors in Parentheses.
***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.
[a]"Informal" refers to informal private sector.
[b]"Formal" refers to formal private sector.
[c]% of individuals that changed the sector = $\frac{\text{No. of obs. treated group}}{\text{Total no. of observation}}$.

to work in the public sector, than people in other countries (Farid, 2007; Barsoum, 2015).[5] Given this state of affairs, we conjecture that the growth of the formal private sector cannot be expected without changing not only the wage and employment systems but also Egyptian culture and ways of thinking. Along with restructuring the wage system and employment, fostering entrepreneurship and changing individual cultures shall be necessary steps to grow (shrink) the formal (informal) private sector.

To this end, some education programs and government policies for restructuring of the labor market or facilitating entrepreneurship in Egypt shall be necessary. In Japan, the policy of "equal pay" is applied by readjusting the wage structure in the public sector to the private sector's wages (Morikawa, 2016). This policy contributes to reducing the wage gaps between the employment sectors and increase the attractiveness of each sector for the new labor market entries. In France, Italy and Spain, the governments make significant efforts to formalize the informal private sector employment by "lowering the labor payroll taxation" and enforcing the legislation of the employment contracts by "increasing inspections" on small businesses (Porto et al., 2016). "Equal pay" policy can be applied in the Egyptian labor market by only incentivizing the private sector to increase wages of employees because of the low wages in that sector. The policy of "lowering the labor payroll taxation" can be implemented in Egypt to reduce seasonal informal private sector employment through reducing taxes for small businesses that give a contract for more than one year. These policies in addition to existing ones can be adopted in other developing countries with readjustment to fit into the cultural and environmental context.

4. Conclusion

We have investigated the wage gap and labor mobility among the public, formal private and informal private sectors with Oaxaca–Blinder decomposition and difference-in-difference methods, using panel ELMPS data that provides information on the Egyptian labor market for three waves, 1998, 2006 and 2012. The results show that the wage gap between the public and formal private (informal private) sectors has consistently remained strong, with education, age

[5] In particular, Barsoum (2015) notes that people in Egypt prefer the public sector jobs to the private sector jobs even if public sector jobs provide lower wages than private sector jobs due to their culture, claiming that working in the public sector has remained and/or will remain the best choice among the available employment sectors in Egypt. During the 1960s, the government applied a policy to guarantee public sector jobs for highly educated individuals. However, this policy was suspended under Egypt's Structural Adjustment Program in 1991 (Assaad, 1997; Elhaddad and Gadallah, 2018). Barsoum (2015) claims that this policy still has an enormous effect on the economy and culture because the public sector is believed by a majority of Egyptian people to be the best employer. It is a sector where highly educated individuals are employed with insurance, social status, paid leaves, pension plans and job security.

and working experience being important driving forces, and the percentages and wage losses of movers from the public sector to the informal private sector are consistently much smaller than the percentages and wage losses of movers from the formal private sector to the informal private sector. Egyptian formal-private sector employees have a higher risk of unwillingly falling into the informal private sector than public sector employees, while the highly educated employees are attracted to and stay long term only in the public sector. Overall, these results suggest likely obstacles and implications for further growth and stability of the Egyptian economy and may be applicable to other developing and Arab countries with a sizable public sector.

Changes in the Egyptian labor market and employment policies appear to be needed. The Egyptian government should restructure the wage system and other benefits associated with employment to obtain a balance between the public and private sectors, providing people with incentive schemes to nurture (formalize) the formal (informal) private sector. Along with such restructuring, we suggest that fostering entrepreneurship as well as changing individual cultures might be necessary steps to grow (shrink) the formal (informal) private sector because a majority of Egyptian people do not pay attention to entrepreneurship, believing that the public sector is the best employer in the economy (Barsoum, 2015). Therefore, implementing educational programs and/or government policies to not only restructure wage and employment systems but also facilitate entrepreneurship among people is recommended. Of course, altering individual culture and attitudes for employment is a large challenge. However, it is our belief that such changes shall be inevitable to achieve the sound growth of the country.

We finally note some limitations and future avenues of research. This research does not fully address cultural and motivational aspects of individual preferences over sectors of employment in the Egyptian labor market. In Egypt or other African countries, social status or acceptance from family, neighbors and friends regarding whether you work in the public or private sector really matters, influencing many events in the future such as marriage. Further data collection and future research should focus on not only the economic factors but also the motivational and cultural factors of how people choose the sectors of employment, which will reveal the necessary policies in terms of the cognitive and cultural aspects of labor markets. We also note that a new wave of ELMPS has been released in November 2019, covering the Egyptian labor market up to 2018. However, it is not included in our analysis because we had witnessed major drastic changes and instability due to politics and revolution in Egypt and these had affected the labor market due to the factors other than economic ones during the period between 2012 and 2018. Thus, future research could analyze the effect of such changes in the labor market of Egypt and how the employment and mobility in different sectors changes. These caveats notwithstanding, we hope that this research will be considered an important first step to understand the problems of the wage gap and labor mobility among sectors in not only Egypt but also other developing countries.

Acknowledgment

The authors thank anonymous referees, Hiroaki Miyamoto, Jingchao Zhang, Khatun Mst Asma, Pankaj Koirala, and Yayan Hernuryadin for their helpful comments, advice and supports. Finally, special thanks to Raja Rajendra Timilsina for his continuous help and encouragement.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Detailed difference-in-difference in log hourly wages

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.eap.2020.09.006.

References

Assaad, R., 1997. The effects of public sector hiring and compensation policies on the Egyptian labor market. World Bank Econ. Rev. 11, 85–118.
Assaad, R., 2014. Making sense of Arab labor markets: The enduring legacy of dualism. IZA J. Labor Dev. 3, 1–25.
Assaad, R., Krafft, C., 2013. The Egypt labor market panel survey: Introducing the 2012 round. IZA J. Labor Dev. 2, 1–30.
Assaad, R., Krafft, C., 2015. The Structure and Evolution of Employment in Egypt. Oxford university press.
Bargain, O., Kwenda, P., 2014. The informal sector wage gap: New evidence using quantile estimations on panel data. Econom. Dev. Cult. Chang. 63, 117–153.
Barsoum, G., 2015. The public sector as the employer choice among youth in Egypt: The relevance of public service motivation theory. Int. J. Public Adm. 39, 205–215.
Blinder, A.S., 1973. Wage discrimination: Reduced form and structural estimates. J. Hum. Resour. 8, 436–455.
Central agency for public mobilization and statistics, 1985. Occupation Codebook. Central agency for public mobilization and statistics.
Cunningham, W.V., Maloney, W.F., 2001. Heterogeneity among Mexico's microenterprises: An application of factor and cluster analysis. Econom. Dev. Cult. Chang. 50, 131–156.
Elhaddad, A., Gadallah, M., 2018. The Informalization of the Egyptian Economy (1998-2012): A Factor in Growing Wage Inequality? Economic Research Forum, Cairo, Egypt, Working paper 1210.
ELMPS, 1998. Egyptian Labor Market Panel Survey 1998. Economic Research Forum, Cairo, Egypt, (www.erfdataportal.com).
ELMPS, 2006. Egypt Labor Market Panel Survey 2006. Economic Research Forum, Cairo, Egypt, (www.erfdataportal.com).

ELMPS, 2012. Egypt Labor Market Panel Survey 2012. Economic Research Forum, Cairo, Egypt, (www.erfdataportal.com).
Elsayed, A., Wahba, J., 2018. Political change and informality: Evidence from the Arab spring. Econ. Transit. Inst. Change 27, 31–66.
Farid, M., 2007. Entrepreneurship in Egypt and the US compared: Directions for further research suggested. J. Manage. Dev. 26, 428–440.
Fernández-de-Córdoba, G., Pérez, J.J., Torres, J.L., 2012. Public and private sector wages interactions in a general equilibrium model. Public Choice 150, 309–326.
Fields, G.S., 1975. Rural–urban migration, urban employment and underemployment, and job search activity in LDCs. J. Dev. Econ. 2, 165–187.
Francisco de Castro, Matteo Salto, H.S., 2013. The Gap Between Public and Private Wages: New Evidence for the EU. Economic papers No. 508.
Gërxhani, K., 2004. The informal sector in developed and less developed countries: A literature survey. Public Choice 120, 267–300.
Glinskaya, E., Lokshin, M., 2007. Wage differentials between the public and private sectors in India. J. Int. Dev. 19, 333–355.
Gunther, I., Launov, A., 2012. Informal employment in developing countries: Opportunity or last resort? J. Dev. Econ. 97, 88–98.
Hammouya, M., 1999. Statistics on Public Sector Employment: Methodology, Structure and Trends. Technical report, International Labour Office.
ILO News, 2018. More than 60 per cent of the world's employed population are in the informal economy. https://www.ilo.org/global/about-the-ilo/newsroom/news/WCMS_627189/lang--en/index.htm. Last checked on Sept 16, 2020.
Lindauer, D., Nunberg, B., 1996. Rehabilitating Government: Pay and Employment Reform in Africa. Technical report, World Bank 13711.
Lucifora, C., Meurs, D., 2006. The public sector pay gap in France, Great Britain and Italy. Rev. Income Wealth 52, 43–59.
Magnac, 1991. Segmented or competitive labor market? Econometrica 59, 165–187.
Maloney, W.F., 1999. Does informality imply segmentation in urban labor markets? Evidence from sectoral transitions in Mexico. World Bank Econ. Rev. 13, 275–302.
Maloney, W.F., 2004. Informality revisited. World Dev. 32, 1159–1178.
Marinakis, A.E., 1994. Public sector employment in developing countries: An overview of past and present trends. Int. J. Public Sect. Manage. 7, 50–68.
Meghir, C., Narita, R., Robin, J.-M., 2015. Wages and informality in developing countries. Amer. Econ. Rev. 105, 1509–1546.
Morikawa, M., 2016. A comparison of the wage structure between the public and private sectors in Japan. J. Japan. Int. Econ. 39, 73–90.
Neumark, D., 1988. Employers' discriminatory behavior and the estimation of wage discrimination. J. Hum. Resour. 23, 279–295.
Oaxaca, R., 1973. Male–female wage differentials in urban labor markets. Internat. Econom. Rev. 14, 693–709.
Oaxaca, R.L., Ransom, M.R., 1994. On discrimination and the decomposition of wage differentials. J. Econometrics 61, 5–21.
Pages, C., Stampini, M., 2009. No education, no good jobs? Evidence on the relationship between education and labor market segmentation. J. Comp. Econ. 37, 387–401.
Papapetrou, E., 2006. The public–private sector pay differential in greece. Public Finance Rev. 34, 450–473.
Porto, E.D., Elia, L., Tealdi, C., 2016. Informal work in a flexible labour market. Oxford Econ. Pap. 69, 143–164.
Pratap, S., Quintin, E., 2006. Are labor markets segmented in developing countries? A semiparametric approach. Eur. Econ. Rev. 50, 1817–1841.
Rama, M., 1997. Efficient Public Sector Downsizing. Technical report, World Bank RPO 679-51.
Reimers, C.W., 1983. Labor market discrimination against hispanic and black men. Rev. Econ. Stat. 65, 570–579.
Rodrik, D., 2000. What drives public employment in developing countries? Rev. Dev. Econ. 4, 229–243.
San, S., Polat, Ö., 2012. Estimation of public–private wage differentials in Turkey with sample correction. Dev. Econ. 50, 285–298.
Wahba, J., 2009. Informality in Egypt: A Stepping Stone or a Dead End. Economic Research Forum, Cairo, Egypt, Working paper 456.